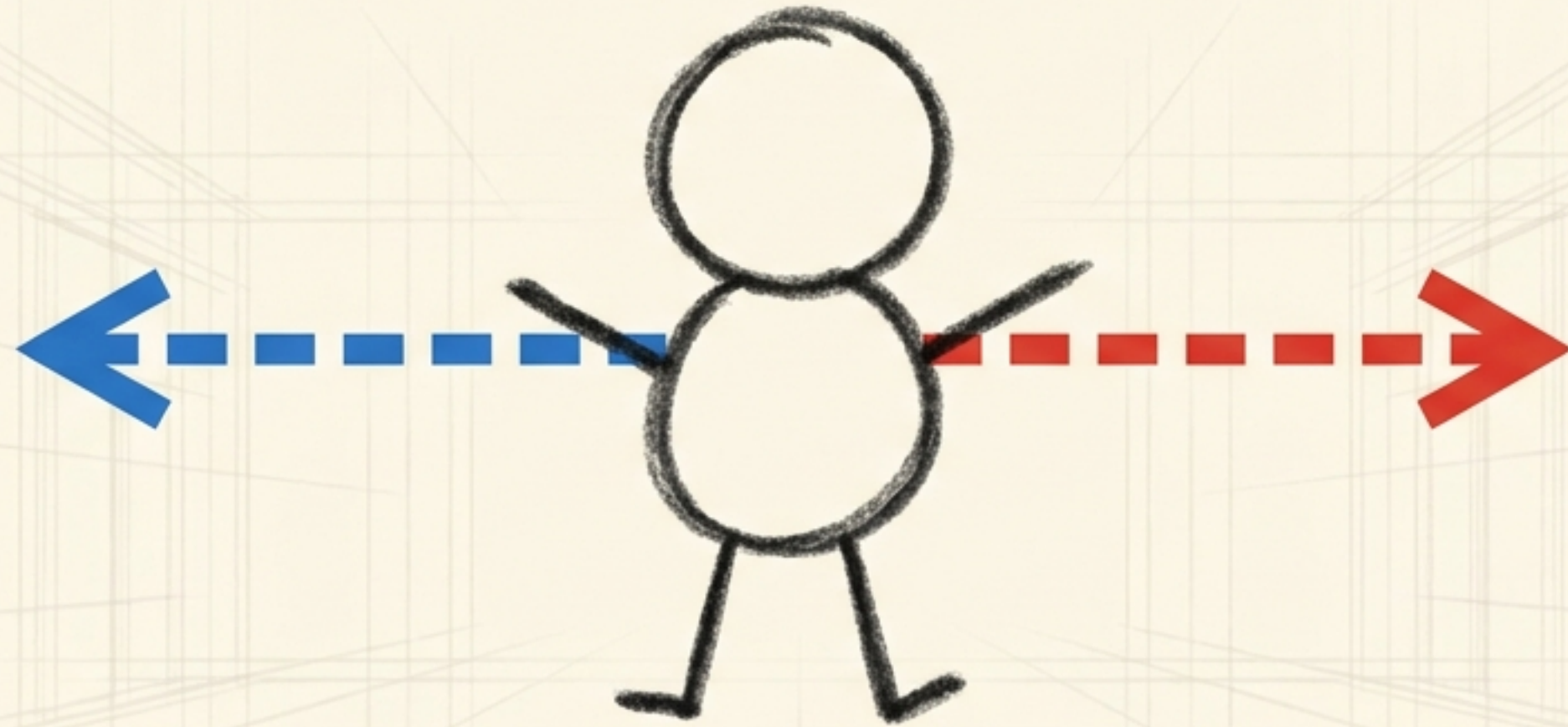


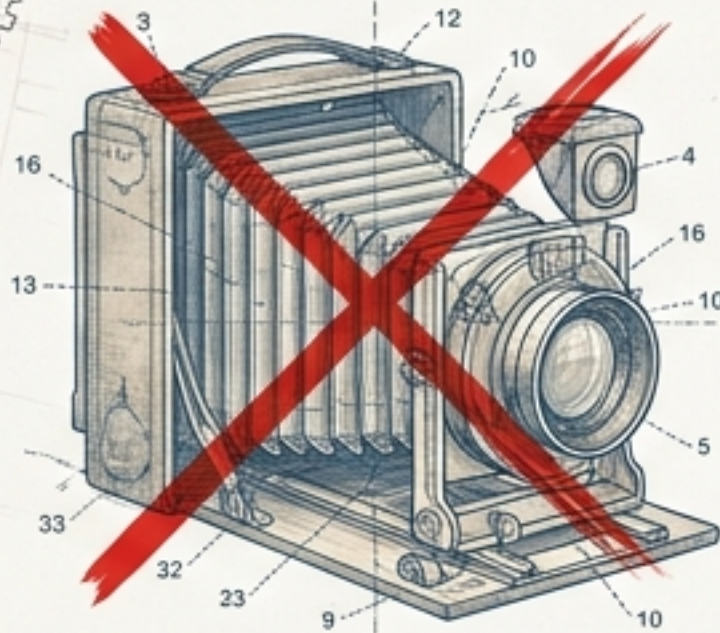
Making Thought Visible

The Two Directions of a Drawing



Synthesizing the cognitive architectures of Jacqueline Goodnow and Judith Fan to reveal how human visual abstraction is organized twice.

A drawing is not a defective photograph



THE RESEMBLANCE TRAP (Luquet)

Drawings were judged by physical realism. Deviations like transparent houses were seen as failures to replicate a single visual viewpoint.



THE DIAGNOSTIC TRAP (Goodenough)

Drawings were treated as symptoms of general intellectual maturity, counting features without regard for how they were procedurally produced.



THE PROJECTIVE TRAP (Machover)

Spatial placement and size proportions were incorrectly read as direct expressions of emotional affect and hidden inner conflict.

Goodnow and Fan reject the finished image as a static deposit of meaning.
A drawing is an organized cognitive achievement.

Every mark faces backward and outward

The Backward Direction

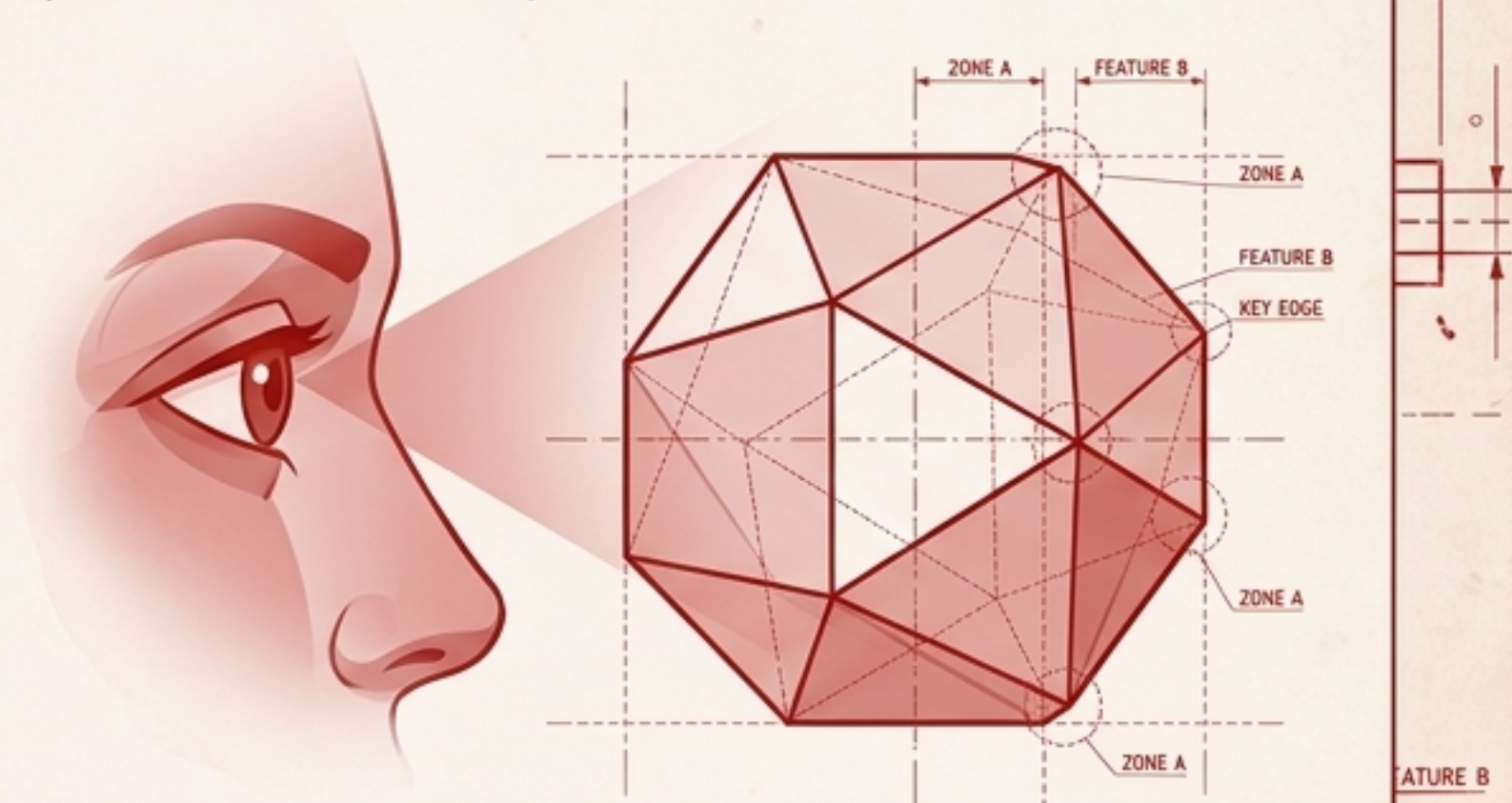
(Jacqueline Goodnow, 1977)



- Focus: Procedure and Process.
- Theory: The drawing is a physical residue of the sequence that produced it. It reveals the syntax of action, spatial constraints, and rules for partitioning space.

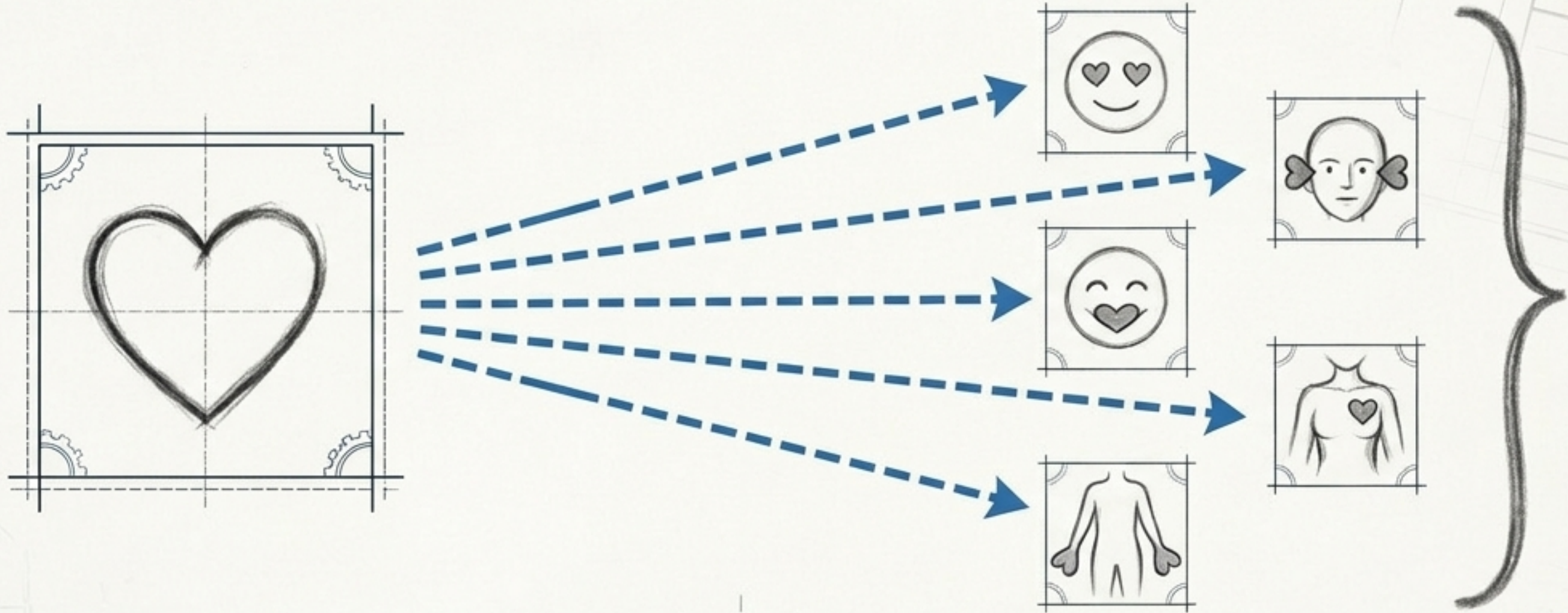
The Outward Direction

(Judith Fan, 2020s)



- Focus: Product and Pragmatics.
- Theory: The drawing is a communicative tool. It reveals how an intended concept is made perceptible to another mind through selective abstraction.

Economy is an achievement of visual thought, not a lack of vocabulary



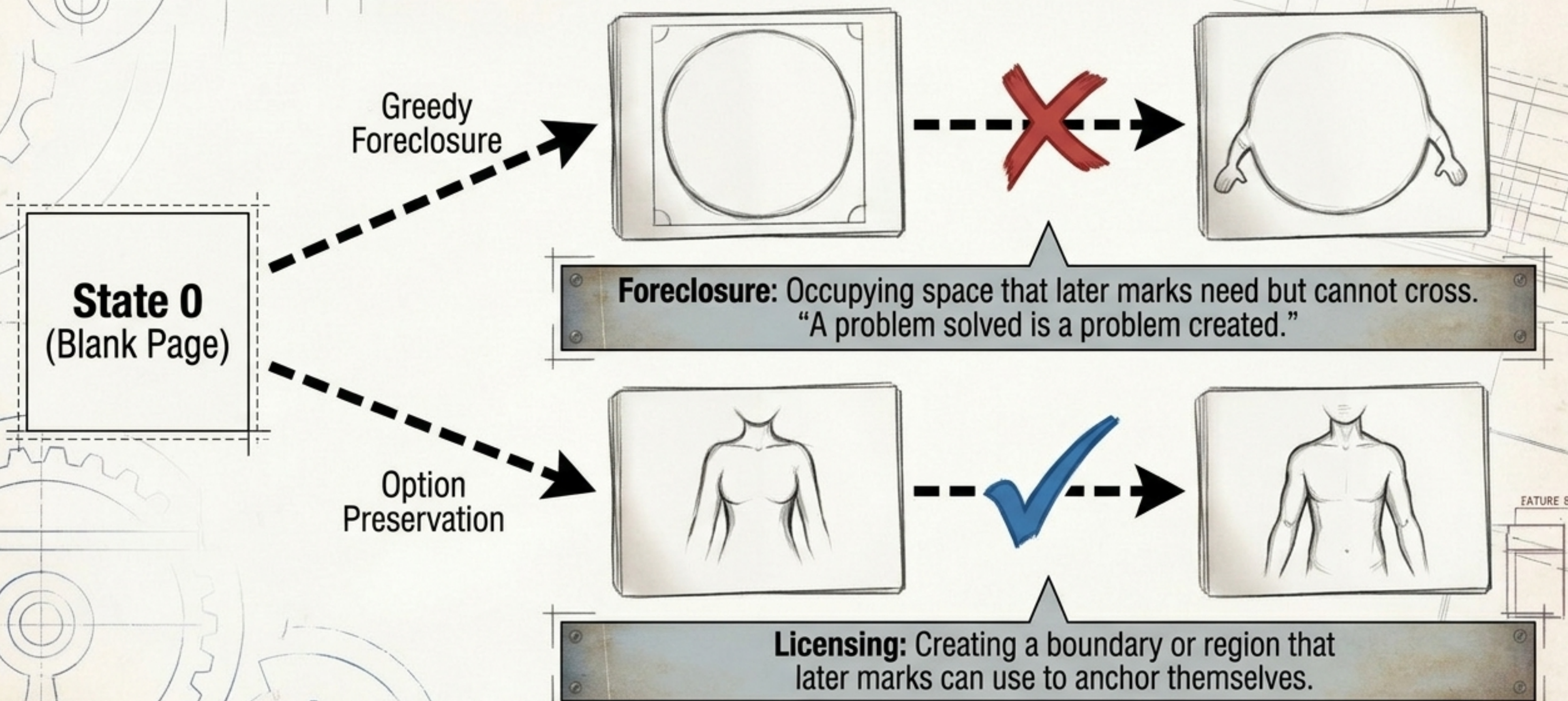
One Unit, Many Uses

A repeated graphic unit is chosen first for its balance and mastered motor form, and only then assigned a semantic role. Form precedes reference.

The Boundary Principle

Young children operate with strict spatial syntax: "To each its own boundary, and to each its own space." Parts are fenced off; overlaps are rare.

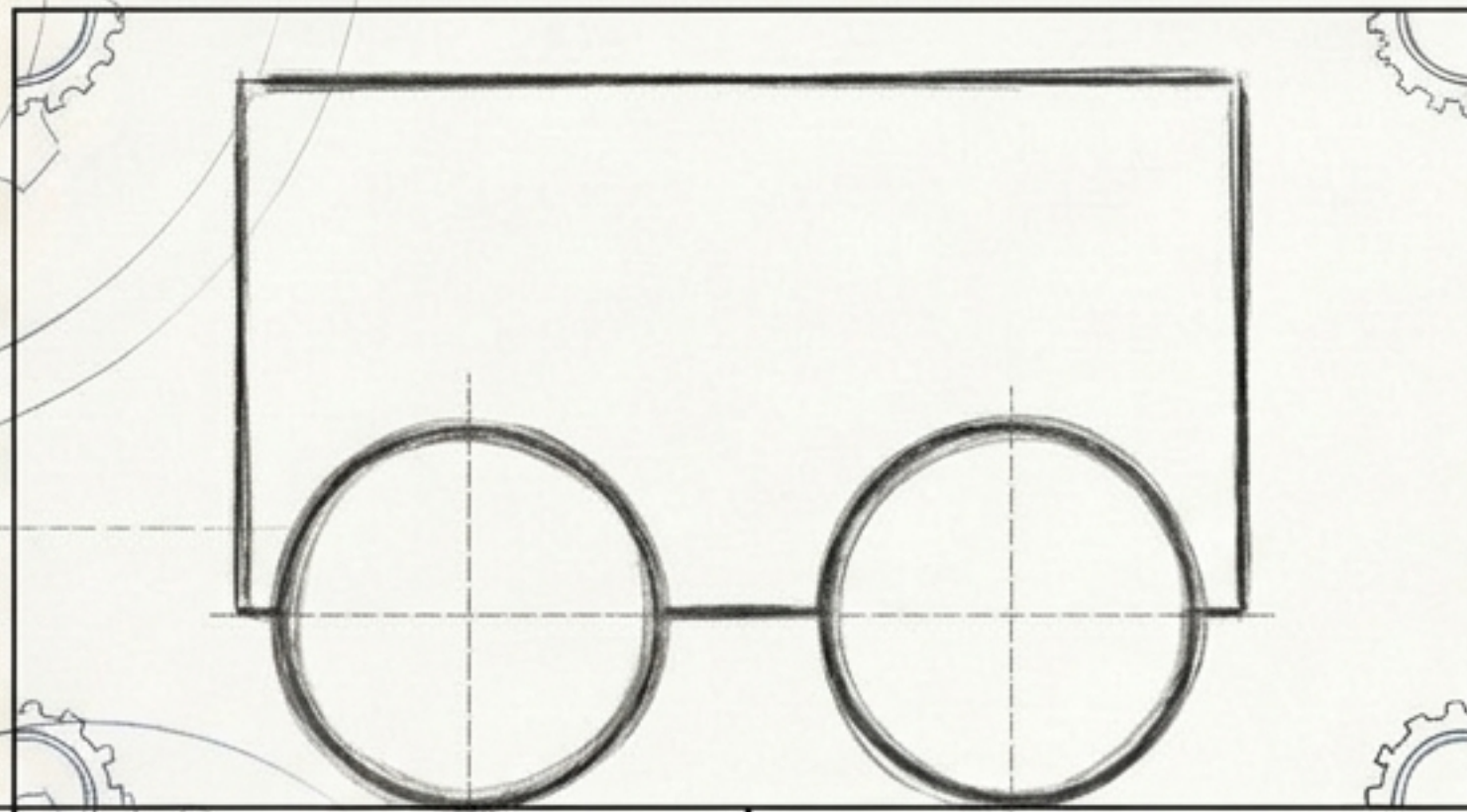
Every stroke changes the future: Licensing and Foreclosure



Marks accumulate without erasure. The images a drawer can reach depend on order, not just intention.

The deviation is the solution: Testing rules with inherited figures

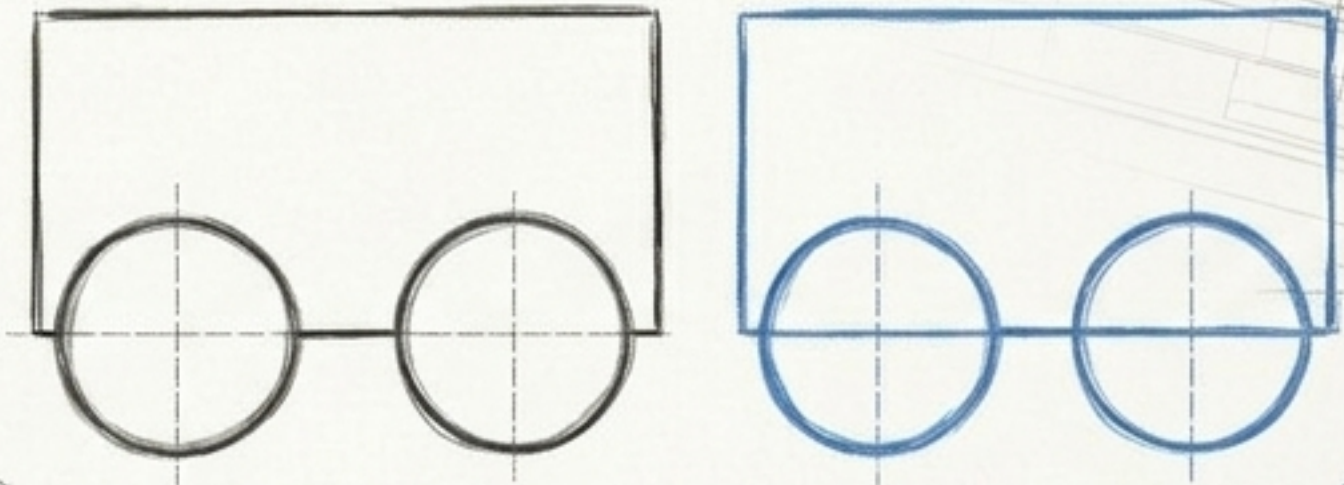
The Prompt



Constraint: Add two more wheels to the train.
Conflict: Must match wheel size and attach to the underside, but doing so requires crossing existing lines (violating the boundary rule).

The Child's Solution

Before & After



The Licensing Move: Instead of overlapping lines, the child draws an entirely new carriage to create new underside space. They redefine the initial constraints to wipe out the difficulty.

Procedural coherence wins over physical fidelity.

Abstraction is an optimization for a specific audience

Fan treats drawing as a massive reduction of reality.
What survives the reduction?

Detailed / Specific



Sparse / Categorical



Contextual Flexibility: Produces faithful sketches when needing to pick out one specific exemplar from a similar lineup.

Shared History: Drawings made for a partner with shared visual history can be exponentially sparser. The shared history carries the meaning.

Depicting an object versus explaining a mechanism

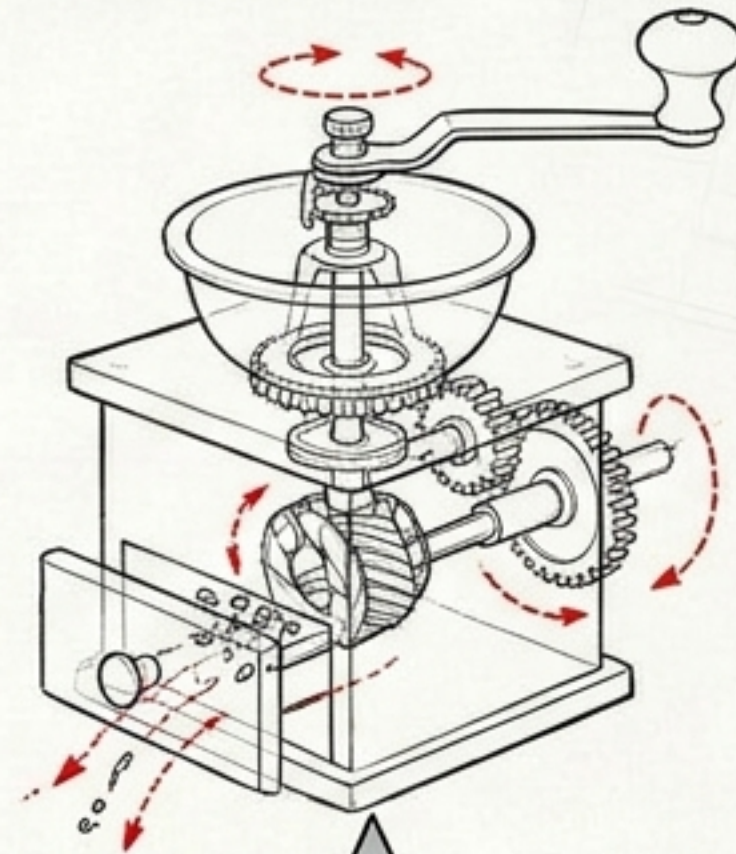
Goal: Depiction



Strategy: Preserves identifying surface features and visual fidelity.

Outcome: Helps the viewer pick out the specific object from a set.

Goal: Explanation

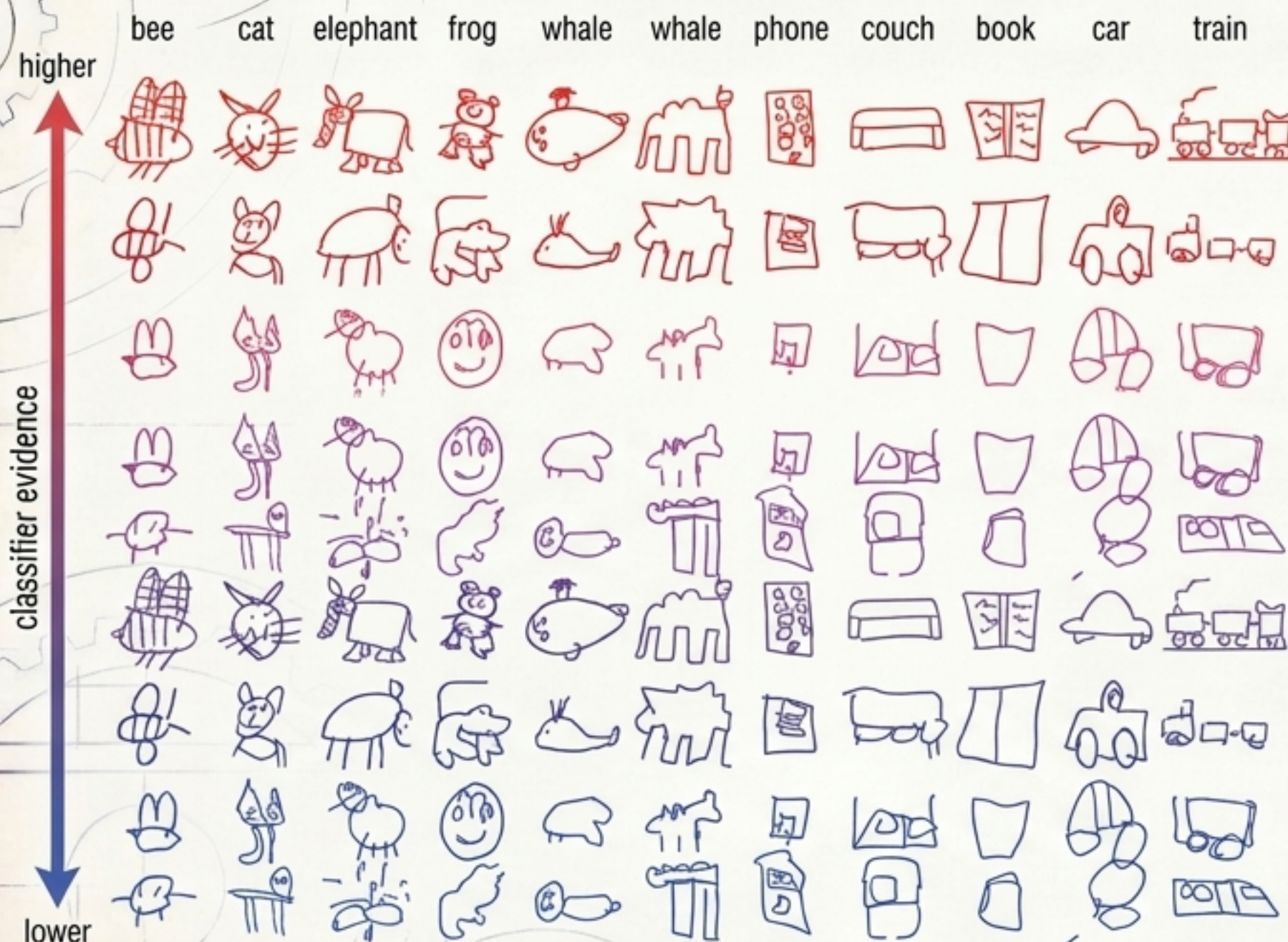


Strategy: Suppresses background detail. Preserves functional relations, causal parts, and motion.

Outcome: Helps the viewer extract mechanistic understanding.

Competent drawing is not a march toward realism;
it is the flexible suppression of irrelevant data.

Information is preserved at multiple grains of resolution



Category-Diagnostic Growth

Analyzing 37,000+ drawings reveals that images become systematically more recognizable with age. Maturation is the increasing presence of information that separates a target category from alternatives.

The Hidden Floor

Even “unrecognizable” drawings at the bottom of the grid systematically retain broad structural properties like animacy and real-world size. Information survives at several grains at once.

Reinterpreting Goodnow

A graphic unit is not just a substituted mark; it is a selective carrier of information tailored to an audience.

The Three Decompositions of a Single Drawing

1. Graphic (Goodnow):
Constructional type
(e.g., "It is an oval").

2. Semantic (Fan):
The part represented
(e.g., "It is a torso").

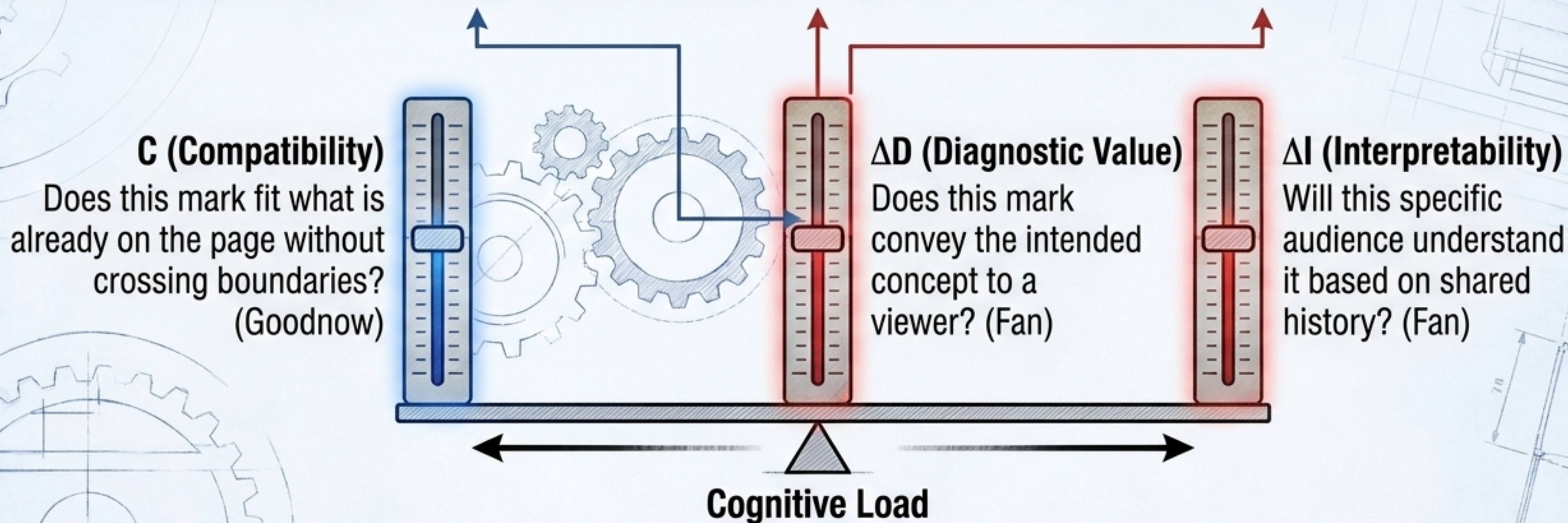
3. Procedural (Goodnow):
Sequence and anchoring
(e.g., "Drawn 3rd,
anchored to the head").

—— Graphic
—— Semantic
-- -- Procedural

Crucial Insight: Goodnow and Fan decompose the same act differently. One graphic unit can serve multiple semantic parts; one semantic part can be built by multiple graphic units.

The Optimization Formula: Every mark is a negotiation

$$V(m \mid s, g, a) = \alpha C(m \mid s) + \beta \Delta D(m \mid s, g) + \gamma \Delta I(m \mid s, a)$$



The Conflict: An explanatory arrow raises diagnostic value (D) but violently violates the non-crossing boundary constraint (C). The drawer must constantly negotiate this trade-off.

The Illusion of the Endpoint

Proposition 2: If two drawers produce the same finished image, endpoint measures judge them as equal. This is a trap.

Human Cognition



Human Cognition: Meaning is preserved through a history of spatial constraints and procedural logic.

The AI Blindspot



The AI Blindspot: Vision models can perfectly generate sketches without sharing human abstraction policies (knowing *what* is expendable and *why*).

Successful recognition does not entail a shared abstraction policy.

Measuring the Invisible Process

Budget Ladders (Fan)

Force drawers to use less ink or time.
The features that survive are the
features prioritized highest.

Inherited-Figure Probes (Goodnow)

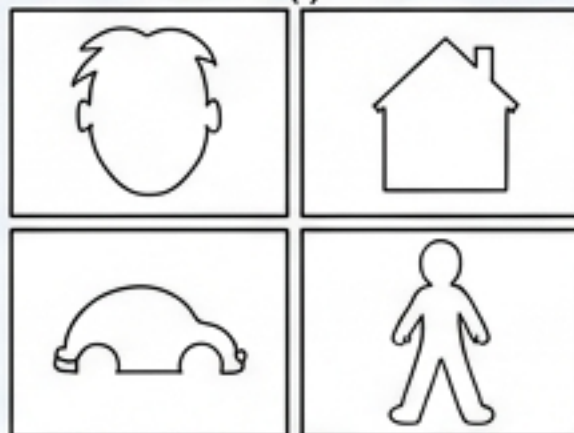
Present a partial figure that blocks
the usual procedural path to expose
spatial rules.

Order Logging

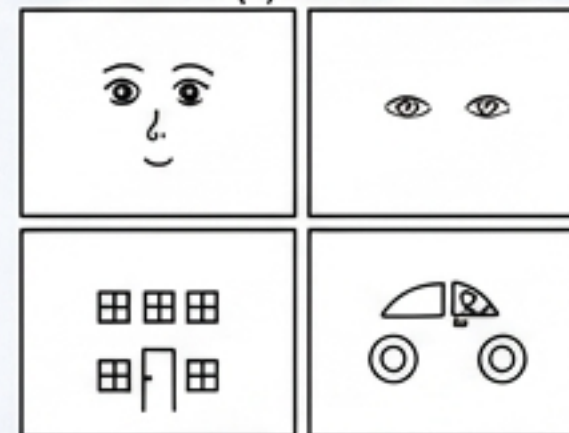
Record the exact sequence to expose
option-preservation heuristics.

Inherited States of Constraints

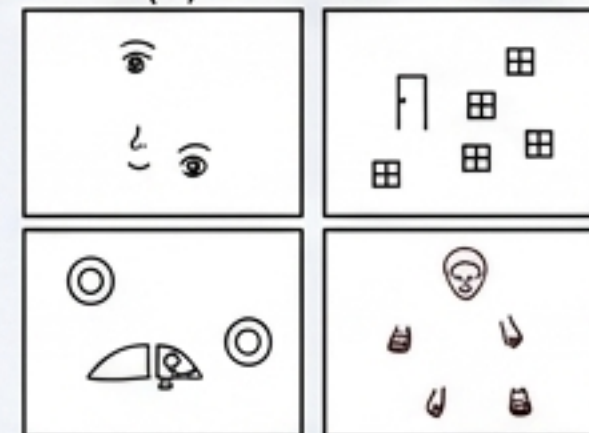
Condition (i): Outline



Condition (ii): Inner features



Condition (iii): Scrambled inner features

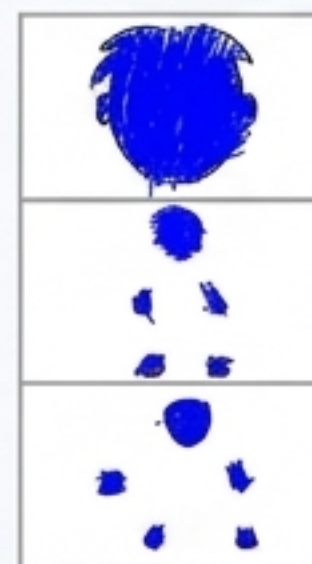


Child Completions (Varied Responses)

Children with high scribbling score



Child coloring in



Child tracing



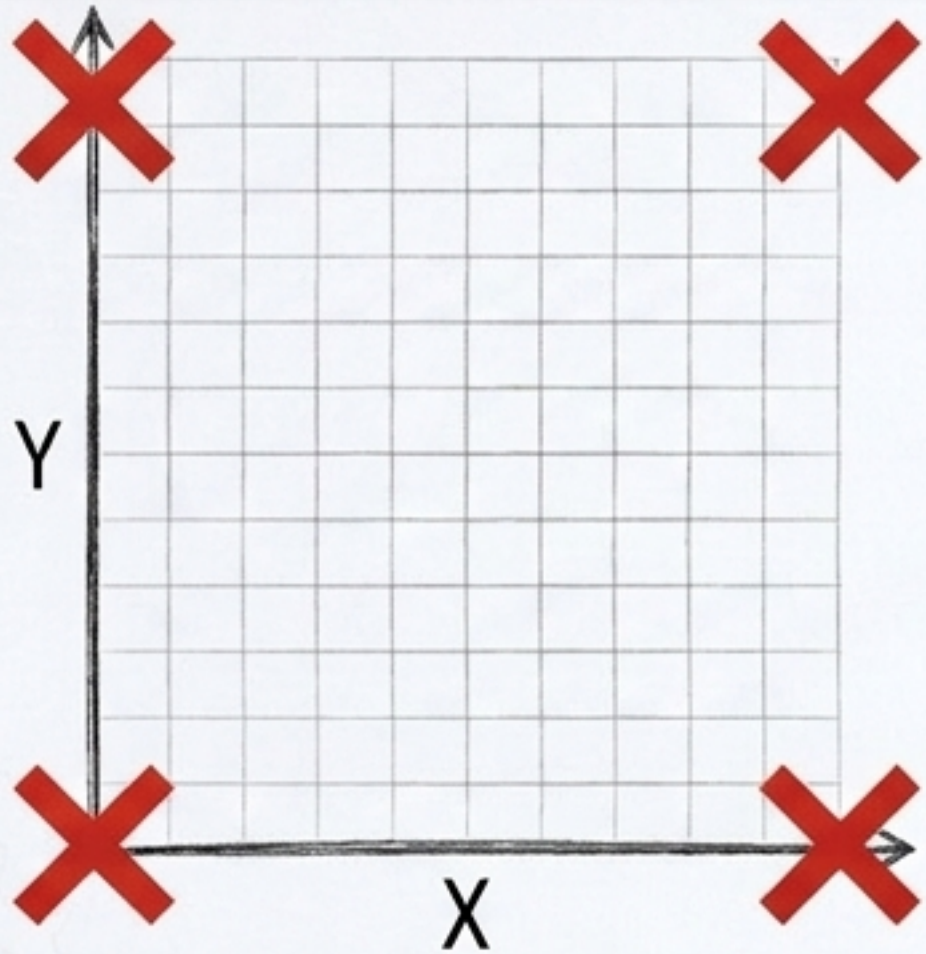
Children with high completion score



To escape endpoint equivalence, we must force the process into the open.

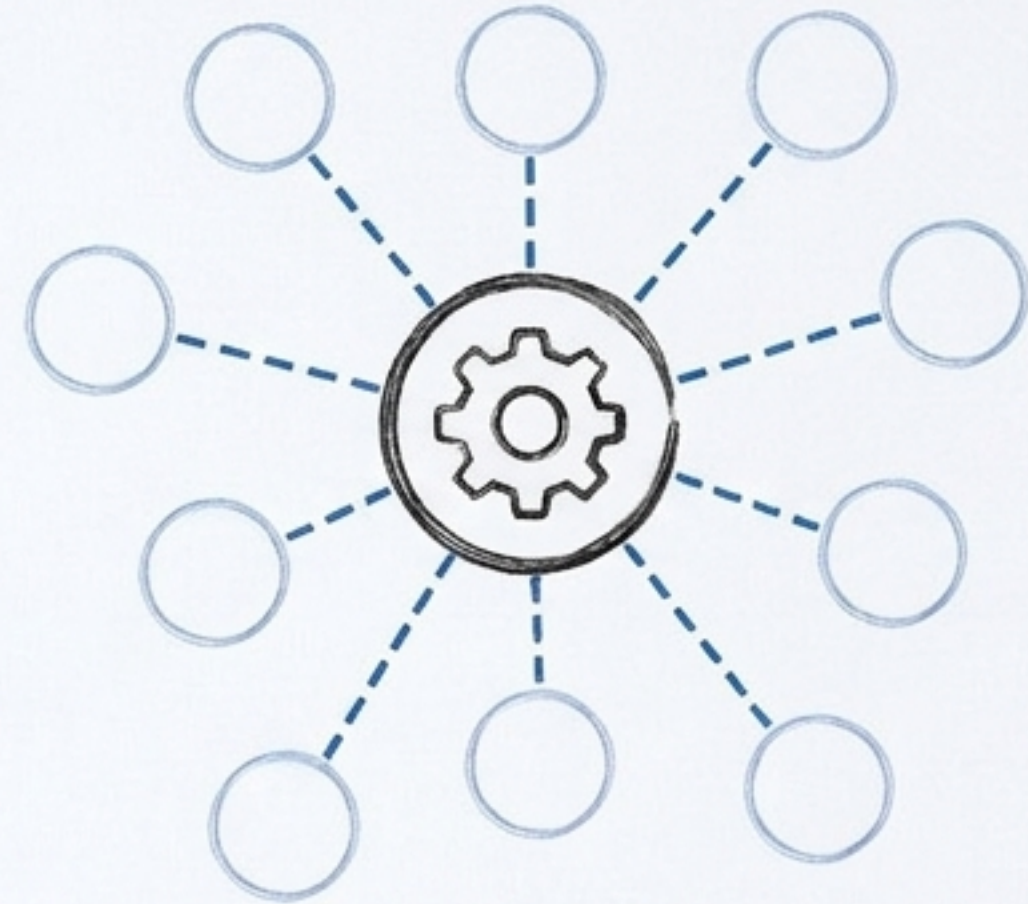
Beyond Pictures: The Architecture of Diagrams and Plots

Sequential Commitment



Choosing the axes of a coordinate plot forecloses some data comparisons before a single data point is even drawn.

Icons of Relations



Placing a core component at the center of a schematic licenses radial connections, but forecloses a linear causal chain.

Diagramming expertise is the mastery of sequencing habits that preserve spatial options for the relations the audience will eventually need.

Visual abstraction is organized twice

Prospectively (The Backward Direction):

Governed by internal rules of syntax, spatial constraints, and which marks can mathematically follow those already made.



Retrospectively (The Outward Direction):

Governed by the distinctions, categories, and mechanisms an external viewer can recover from the final image.

A complete theory of visual abstraction cannot just measure if a drawing communicates; it must understand the history of constraints that made it possible.